# Blinatumomab (Blincyto®)

Blinatumomab (Blincyto®) is a cancer drug. It is used to treat acute lymphoblastic leukaemia (ALL).

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## What is blinatumomab (Blincyto®)?

Blinatumomab is also called Blicyto®. It is a type of targeted therapy drug called a monoclonal antibody. It is also called a bispecific antibody.

Blinatumomab works by targeting a certain protein on leukaemia cells (CD19). It also targets a protein on the immune cells (CD3). This encourages the immune system to recognise, attack and kill the leukaemia cells.

Blinatumomab is used to treat a specific type of leukaemia called B-cell acute lymphoblastic leukaemia (ALL).

It is best to read this information with our general information about targeted therapy drugs and the type of cancer you have.

Your cancer doctor will talk to you about this treatment and its possible side effects before you agree (consent) to have treatment.

### More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the electronic Medicines Compendium (eMC) website, which has patient information leaflets (PIL) for individual drugs.

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# How blinatumomab is given

You usually have blinatumomab during a stay in hospital. You may have blinatumomab through an infusion pump at home.

During a course of treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During your course of treatment, you will have regular blood tests. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your targeted therapy.

You have blinatumomab as a drip into a vein (intravenous infusion).

Your nurse will give you blinatumomab through 1 of the following:

- a central line – a fine tube that goes under the skin of the chest and into a vein close by
- a PICC line – a fine tube that is put into a vein in the arm and goes up into a vein in the chest
- an implantable port (portacath) – a disc that is put under the skin on the chest or arm and goes into a vein in the chest
- a cannula – a short, thin tube the nurse puts into a vein in the arm or hand.

## Your course of treatment

Blinatumomab is given in cycles of treatment. Each cycle takes 42 days (6 weeks). You have a continuous drip (infusion) of blinatumomab for 28 days (4 weeks). This is followed by a break from treatment for 14 days (2 weeks).

You may have up to 5 cycles of blinatumomab. The number of cycles you have will depend on:

- how the cancer responds to treatment
- any side effects you have.

Before starting blinatumomab, you have a steroid called dexamethasone. You usually have this as a slow injection (bolus) into a vein. You may also be given paracetamol tablets for the first 2 days of the infusion. This is to prevent a reaction to blinatumomab. Some people may need to stay in hospital for the first few days of treatment. This is to monitor any side effects. Your doctor or nurse will let you know if you need to do this.

You usually have the blinatumomab infusion through a small portable pump. You can carry the pump on a belt or in a bag.

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Some people may be allowed to go home after a few days to continue the infusion at home. If you are allowed to go home, your nurses will tell you how to look after the pump. You may need to come back to the hospital every few days to have the infusion changed.

Your doctor, nurse or pharmacist will discuss your treatment plan with you.

Your cancer team may also give you other medicines to take home. Take all your capsules or tablets exactly as they tell you to.

# About side effects

## About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

## Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

## Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

## More information about this treatment

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# Side effects while treatment is being given

Some people may have side effects while they are having this treatment or shortly after they have it.

## Allergic reaction

Some people have an allergic reaction while having this treatment. The first infusion is the most likely to cause a reaction, so it is usually given more slowly than later treatments. Before treatment, you will have medicines to help prevent or reduce any reaction.

Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your infusion. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

# Very common side effects

## Common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

## Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an infection. A low white blood cell count is called neutropenia.

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An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

**It is important to follow any specific advice your cancer treatment team gives you.**

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

## Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

## Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
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breathless

- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

## Effects on the nervous system

Blinatumomab can affect the nervous system. Symptoms include:

- headache
- feeling shaky
- numbness or tingling
- problems with speaking or movement
- seizures (fits)
- confusion
- memory problems
- feeling unsteady or dizzy
- feeling drowsy.

**Tell your doctor or nurse straight away if you have any of these symptoms. They may make some changes to your treatment.**

If you have these side effects, it is important not to drive or operate machinery.

## Cytokine release syndrome (CRS)

This treatment can cause the immune system to release proteins called cytokines into the body. High levels of cytokines may cause inflammation throughout the body. This is called cytokine release syndrome (CRS).

Symptoms of CRS include:

- a high temperature (fever)
- flu-like symptoms such as headache, chills and muscle and joint pain
- low blood pressure and dizziness
- diarrhoea
- rash
- a fast heart rate
- shortness of breath
- swelling caused by a build-up of fluid.

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Tell your doctor or nurse straight away if you have any of these symptoms. They will stop the blinatumomab infusion and treat any symptoms you have. You may need steroids and other treatments.

## Feeling sick

You may feel sick during this treatment, but this is usually mild. The nurses will give you anti-sickness drugs regularly. This will help prevent or control sickness during your treatment in hospital. If you are at home and you feel sick or are sick (vomit), tell your doctor, nurse or pharmacist straight away. They can change the anti-sickness treatment. Or they might change the way you have it to something that works better for you.

## Constipation

This treatment can cause constipation. Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- Eat high-fibre foods, such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

## Diarrhoea

This treatment may cause diarrhoea. Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

## Tummy pain

You may get pain in your tummy (abdomen), or have indigestion. Your doctor, nurse or pharmacist can give you advice or treatment to help.

Contact the hospital straight away on the 24-hour number if you have these symptoms during treatment or after it finishes:

- a sharp pain in your upper tummy (abdomen)
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sickness and vomiting.

### Skin changes

This treatment can affect your skin. It may cause a rash, which may be itchy. Your doctor or nurse can tell you what to expect. If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

Always tell your doctor or nurse about any skin changes. They can give you advice and may prescribe creams or medicines to help. Changes to your skin usually improve when treatment ends.

### Effect on your blood pressure

Blinatumomab can affect your blood pressure. It may cause either low or high blood pressure. If you already have problems with your blood pressure, let your doctor know.

### Back or bone pain

You may get pain in your back or bones around the body, including your hands and feet. If this happens, tell your doctor so they can give you painkillers. Tell them if the pain does not get better. Having warm baths and resting regularly may help.

### Liver changes

Blinatumomab may affect the liver. This is usually mild and goes back to normal after treatment finishes. You will have regular blood tests to check how well your liver is working.

## Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

### Tumour lysis syndrome (TLS)

Some people are at risk of developing a condition called tumour lysis syndrome (TLS). The risk is highest at the beginning of treatment. TLS happens when treatment makes large numbers of cancer cells die and break down quickly. This releases lots of waste products into the blood and can affect the kidneys and heart.

TLS can be prevented or treated. You will have regular blood tests to check for TLS. You may have:

- extra fluids through a drip
- medicines such as allopurinol tablets or rasburicase through a drip.

Drinking at least 2 litres (3½ pints) of fluid a day will also help.

### Effects on the lungs

This treatment can cause changes to the lungs. Tell your doctor, nurse or pharmacist if you develop:

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- a cough that does not go away
- wheezing
- breathlessness.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

### Difficulty sleeping (insomnia)

Sometimes this treatment can affect your sleep. Tell your doctor, nurse or pharmacist if cancer treatment makes it difficult to sleep.

## Other important information

### Blood clot risk

Cancer and some cancer treatments can increase the risk of a blood clot. Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

### Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- herbal drugs and complementary or homeopathic therapies
- recreational drugs – for example, cannabis.

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## Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and coronavirus (covid). They may also recommend other vaccines, such as Shingrix® for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

## Contraception

Your doctor, nurse or pharmacist will advise you not to get pregnant or make someone pregnant while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

## Breastfeeding

You are advised not to breastfeed while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

## Fertility

Some cancer drugs can affect whether you can get pregnant or make someone pregnant. If you are worried about this, it is important to talk with your doctor before you start treatment.

## Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

# About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

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## References

Visit the electronic Medicines Compendium (eMC) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

Cancer Information Development team

Author

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Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

## The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

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# How we can help

## Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.

### Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

### Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

### Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

### Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

### Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

**Get help**

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